

AMENDMENT NO. 1 NOVEMBER 2014
TO
IS 4326 : 2013 EARTHQUAKE RESISTANT DESIGN
AND CONSTRUCTION OF BUILDINGS - CODE OF PRACTICE

(Third Revision)

(Cover page, Foreword, para 5) — Insert the following new para:

‘For provisions on confined masonry building construction, see Annex A.’

(Page 32, Annex A) — Renumber the existing Annex A as Annex B and insert the following as Annex A:

ANNEX A
(Foreword)
CONFINED MASONRY BUILDING CONSTRUCTION

A-1 UNDERSTANDING CONFINED MASONRY CONSTRUCTION

A-1.1 Confined masonry construction is a building technology that offers an alternative to ‘Minimally reinforced masonry’ with ‘RC Bands and Vertical Bars’ and the ‘RC Frame Construction’. It consists of masonry walls (made either of clay brick or concrete block units) and horizontal and vertical RC ‘confining members’ built on all four sides of the masonry wall panels. Vertical members are called ‘tie-columns’ or ‘practical columns’ and though they resemble columns in RC frame construction but are of much smaller cross-section. Horizontal elements, called ‘tie-beams’, resemble beams in RC frame construction, but of much smaller section. It must be understood that the confining elements are not beams and columns the way these are used in RC Frames but rather in the nature of horizontal and vertical ties or bands for resisting tensile stresses and may better be termed as such.

The structural components of a building using confined masonry walls are as follows (*see Fig. 35*):

- a) *Masonry walls*, are load bearing elements, and transmit the gravity loading from the slab(s) and walls above down to the foundation. The walls also work as bracing panels acting along with the confining tie elements which resist the horizontal earthquake forces.
- b) *Confining elements*, horizontal and vertical tie elements provide the necessary tensile strength and ductility to the masonry wall panels and protect them from disintegration in the specified major earthquakes.
- c) *Floor and roof slabs*, transmit both vertical gravity and lateral loads to the confined masonry walls. In an earthquake the slabs behave like rigid horizontal diaphragms.
- d) *Plinth band or tie-beam*, shall transmit the vertical and horizontal loads from the walls down to the foundation. It also protects the ground floor walls from settlement in soft soil conditions.
- e) *Foundation*, transmits the loads from the structure to the ground.

The masonry wall construction shall follow all detailed specifications given in this standard except those to be modified as per this Annex.

Confined masonry walls can be constructed using different types of masonry units, such as hollow clay tiles, burnt clay bricks, concrete blocks of hollow or solid types or dressed rectangular stones. In confined masonry, the reinforcement is concentrated in vertical and horizontal confining elements whereas the masonry walls are usually free of reinforcement, but may be connected with the confining elements using steel dowels.

A-1.2 Difference between Confined Masonry and RC Frame Construction

The appearance of finished confined masonry construction and a RC frame construction with masonry infills may look alike but in load carrying scheme they are very different. The main differences are related to the construction sequence, as

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well as to the manner in which these structures resist gravity and lateral loads (*see* Fig. 36). Whereas in RC Frames, the RC columns and beams carry the vertical gravity as well as the lateral loads from earthquakes or wind storms unaided by the masonry infills, in the case of Confined masonry buildings, the wall panels are the main load carrying elements (both vertical and horizontal) aided by the confining elements for resisting tensile forces.

A-2 GUIDE TO EARTHQUAKE-RESISTANT CONFINED MASONRY CONSTRUCTION

The satisfactory earthquake performance of confined masonry is due to the joint action of masonry walls and the reinforced concrete confining elements. Properly designed and built confined masonry buildings are expected to exhibit good performance even in Maximum Considered Earthquake (MCE) wherein, moderate cracking in the elements is not ruled out, but collapse of building will be highly improbable. Depending on the crushing strength of the building unit, confined masonry buildings may be constructed upto five storeys in height for various Building Categories (7.1) as suggested below:

- a) Categories B and C : Upto 5 storeys
- b) Categories D and E : Upto 4 storeys

A-2.1 Building Configuration

The architectural configuration concepts as highlighted in 4.4 are necessary in confined masonry construction also.

A-2.2 Confining Elements

The tie-beams should be placed at plinth and every floor level. Vertical spacing of tie-beams should not exceed 3 m. The tie-columns should be placed at a maximum spacing of 4 m in 200 mm or thicker walls and 3 m in 100-114 mm thick walls, as well as at the following locations:

- a) At the corners of rooms and all wall-to-wall intersections,
- b) At the free end of a wall, and
- c) At the jambs of doors / windows of 900 mm or wider openings.

A-2.3 Walls

The wall thickness may be kept 100 or 114 mm in the case of one to two storey high residential buildings. But for all important (*see* 6.4.2 of IS 1893) buildings and those of more than two storeys height, the thickness should be 200 or 230 mm, and the mortar shall be as per 8.1.2. At least two fully confined panels should be provided in each direction of the building. The earthquake performance of a confined masonry building depends on the shear resistance of masonry walls. Therefore, it is essential to provide an adequate number of walls in each direction. The walls should be placed preferably at the periphery so as to minimize torsion of the building in an earthquake.

A-2.4 Wall Density

Wall density can be defined as the total cross sectional areas of all confined wall panels in one direction divided by the sum of the floor plan areas of all floors in a building. Wall density of at least 2 percent in each of two orthogonal directions is required to ensure good earthquake performance of confined masonry building in Seismic Zone III of India. For the aim of achieving adequate earthquake performance in the higher Seismic Zones, the wall density of 3 percent in Zone IV and 4.5 percent in Zone V should at least be used.

A-3 CONSTRUCTION DETAILS OF CONFINED MASONRY

A-3.1 Construction of walls

The aim should be to use good quality building materials, that is, the build units and the mortar; as well as good quality workmanship as follows:

- a) Minimum wall thickness is 100 or 114 mm. Wall panel height to thickness ratio should not exceed 30.

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- b) Toothed edges should be left on each side of the wall; the tooth projection may be kept ≤ 40 mm to achieve full concrete filling in the teeth space. Use of horizontal dowels instead or in addition to teething could be made at the wall-to-column interface.
- c) Concrete is to be poured in the tie-columns upon completion of desirable wall height [see Fig. 36(b)]. Bricks must be wetted before casting of concrete.
- d) Formwork support must be provided on two sides of the wall (see Fig. 37). The concrete needs to be vibrated to fill the teeth space thoroughly.

A-3.2 Tie-columns

- a) The minimum tie-column cross sectional dimensions are 100 mm by 100 mm or 114 mm $\times$ 114 mm in 100 or 114 mm thick walls respectively and 150 $\times$ 200 or 150 $\times$ 230 in 200 or 230 mm thick walls at intermediate points but of square section of 200 or 230 mm side at the corners.
- b) Reinforcement for the ground storey tie-columns should be assembled before the foundation construction takes place (see Fig. 38).
- c) The reinforcement in tie-columns at the corners should consist of four high strength deformed bars (HSD/TOR) for longitudinal reinforcement, and 6 mm ties at 100 mm spacing in the end 500 mm height at top and bottom of column and 200 mm apart in the remaining height. Vertical bars should be lapped by a minimum of 50 times the longitudinal bar diameter. The longitudinal bar may be kept as follows:

Categories B and C : 8 mm dia upto four storeys and 10 mm dia in 5 storey building
Categories D and E : 10 mm dia upto 3 storeys and 12 mm dia in 4 storeys building

- d) The tie-columns at jambs of windows / doors may be 100 mm thickness of wall in size and have only two longitudinal bars each of same dia as in corner columns with 6 mm link ties spaced as in the corner columns.

A-3.3 Tie-beams

- a) The minimum tie-beam cross-sectional dimensions are 100 x 100 mm in 100 mm thick walls; 150 $\times$ 200 in 200 mm thick walls and 150 $\times$ 230 in 230 mm thick walls.
- b) Tie-beams are constructed on top of the walls at each floor level.
- c) The tie-beam reinforcement should consist of four high strength deformed bars (HSD/TOR) for longitudinal reinforcement, and 6 mm stirrups at 200 mm spacing. The tie-beam reinforcement needs to be continuous, with longitudinal reinforcement bars lapped by at least 50 times the bar diameter. The following bar diameter may be adopted:
 - 1) Building Categories Band C : 8 mm diameter
 - 2) Building Categories D and E : 10 mm diameter
- d) The lintel level bands may be provided as per 8.4.5. Very wide windows may require stronger lintels.
- e) Proper detailing of the tie-beam and its connection to *tie-column* is a must for satisfactory earthquake performance (see Fig. 39).

A-3.4 Foundation and plinth construction

A plinth band should be constructed on top of the foundation (same as in 8.4.6). Instead of the specified band in this standard, a tie beam as specified in A-3.3 above shall be preferable for proper confinement of masonry panel above it.

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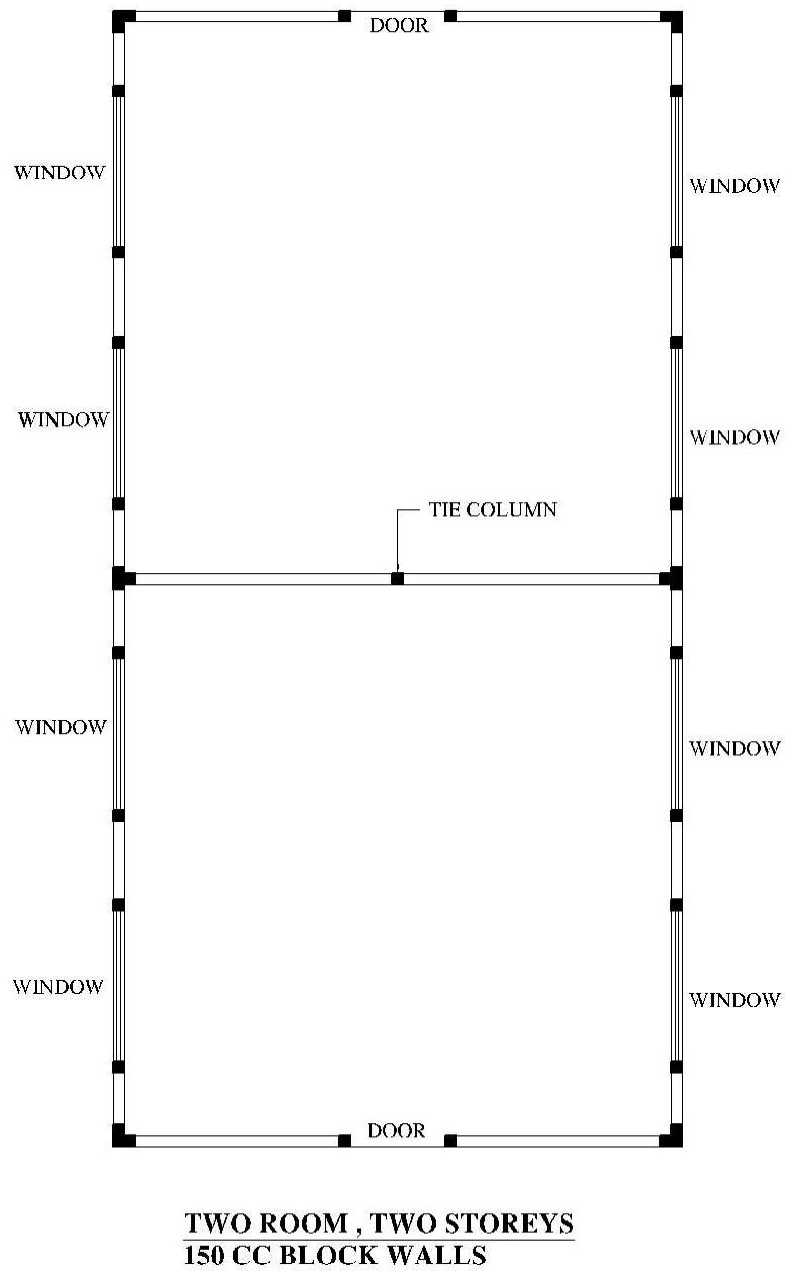


FIG. 35 (a) TYPICAL CONFINED MASONRY BUILDING PLAN

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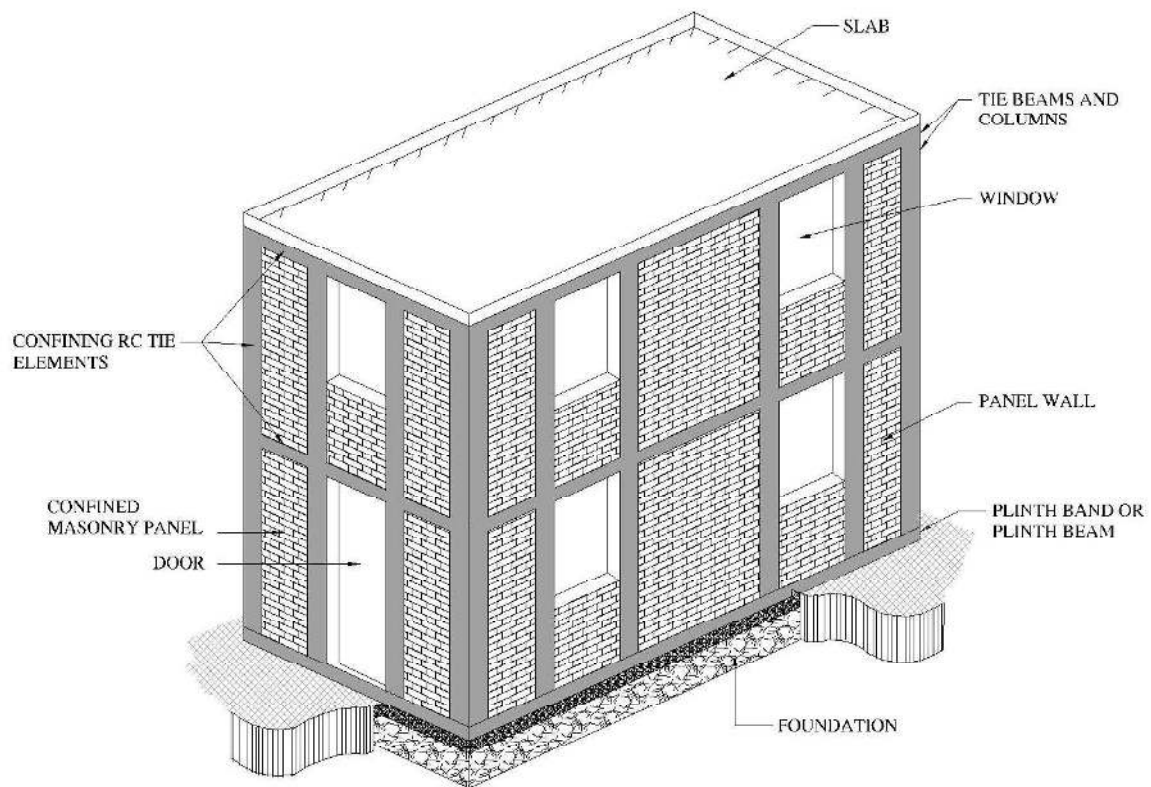


FIG. 35 (b) A TYPICAL 2-STOREY CONFINED MASONRY BUILDING

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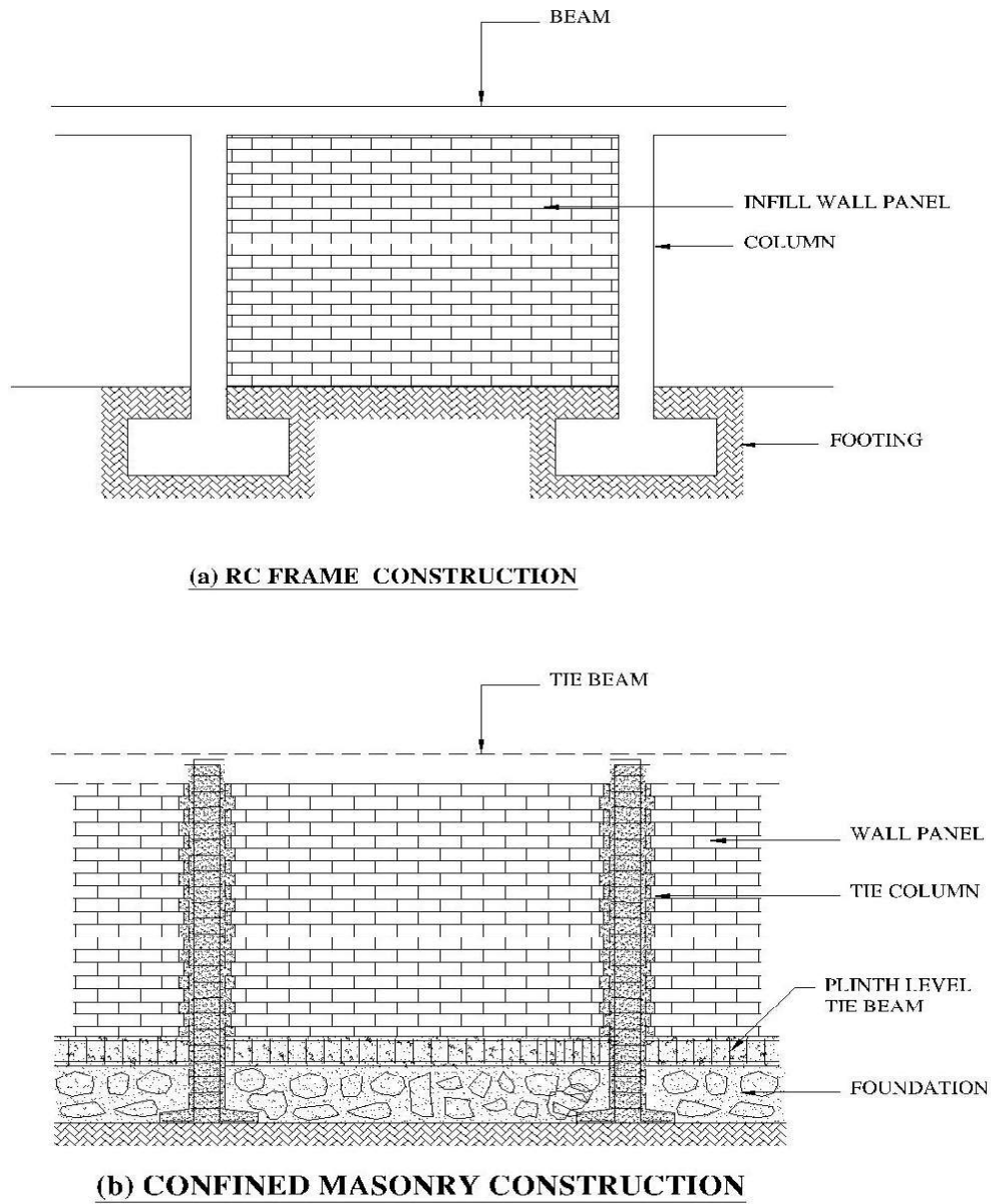


FIG. 36 RC FRAME AND CONFINED MASONRY CONSTRUCTION

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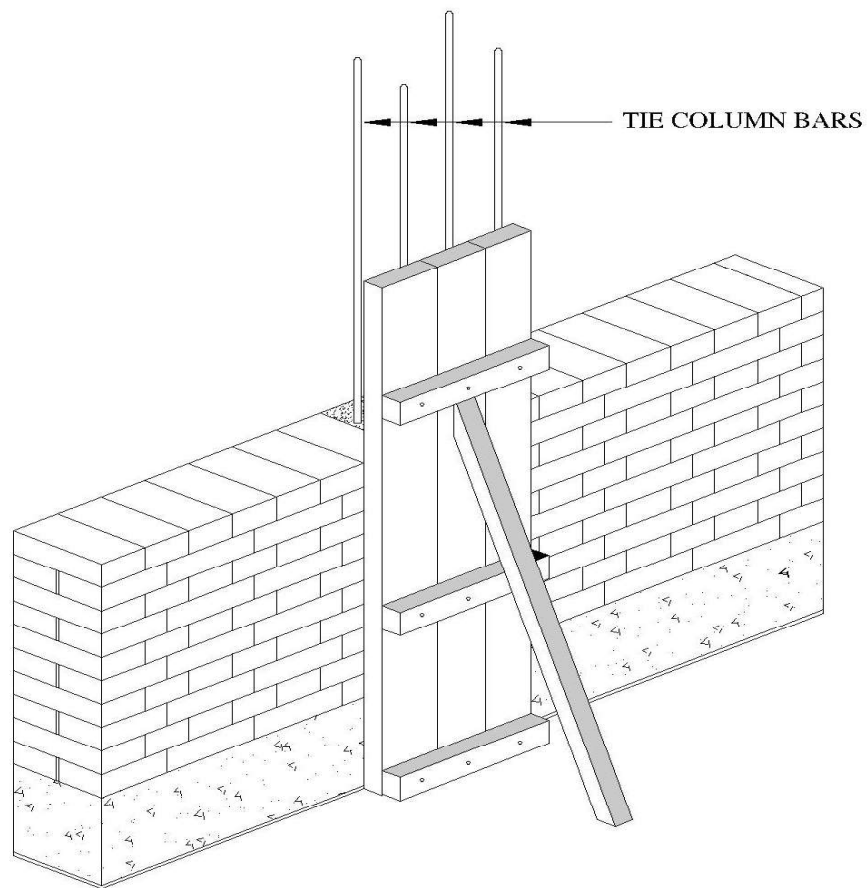


FIG. 37 FORMWORK FOR TIE COLUMNS

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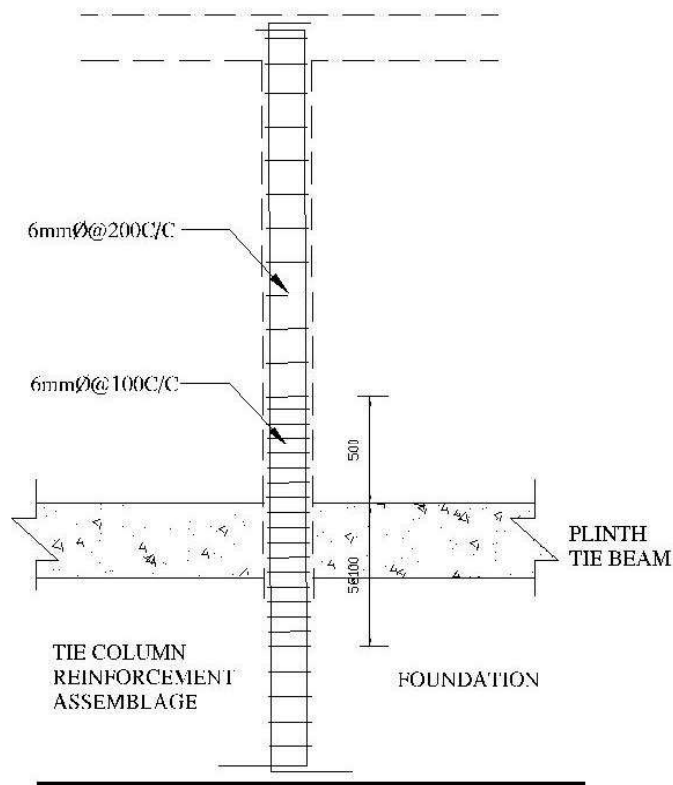


FIG. 38(a) TIE COLUMN REINFORCEMENT SET IN POSITION PRIOR TO THE CONSTRUCTION OF FOUNDATION AND WALLS

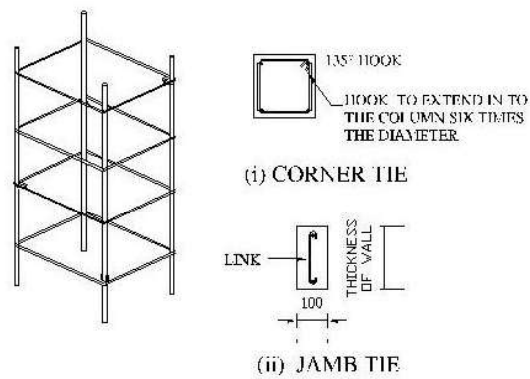


FIG. 38(b) REINFORCEMENT DETAILS OF TIE COLUMN

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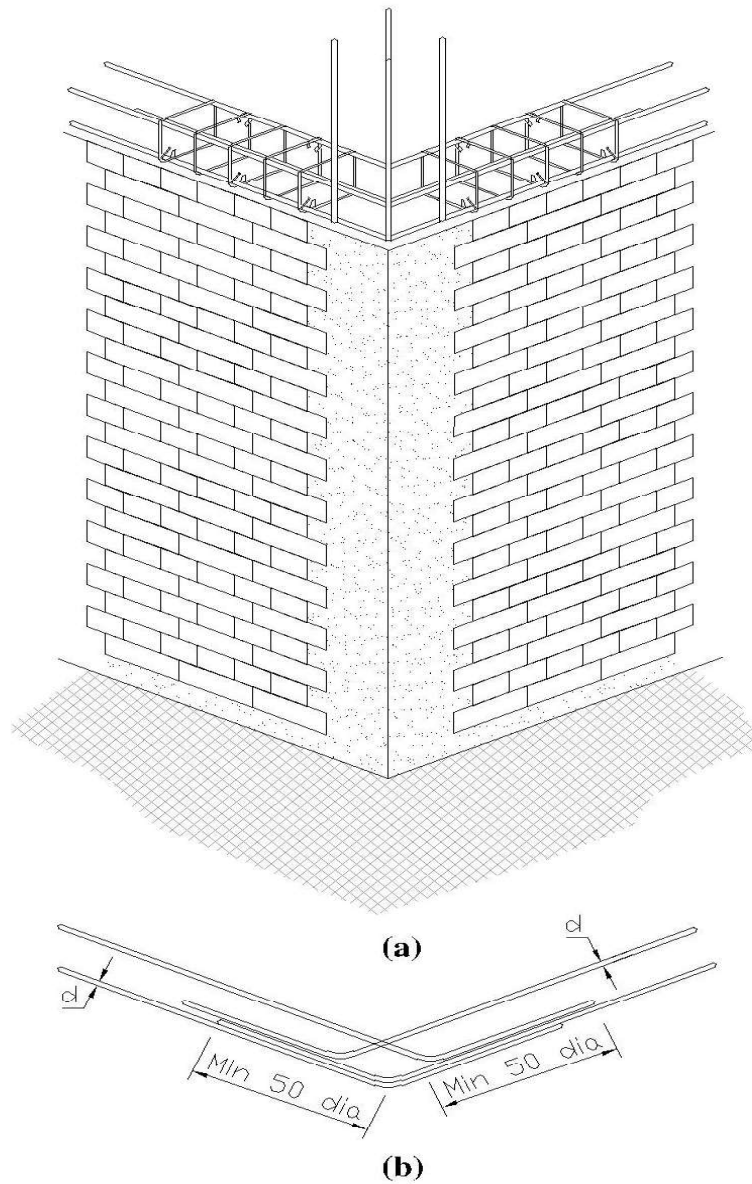


FIG. 39 TIE-BEAM AND COLUMN